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Name: _____

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Department/School: _____

SID: _____

University Physics

Final Examination

Kuang Yaming Honors School, Nanjing University

$$\Gamma(\nu) = \int_0^{+\infty} e^{-x} x^{\nu-1} dx, \quad \Gamma(\nu+1) = \nu\Gamma(\nu), \quad \Gamma(1) = 1, \quad \Gamma(1/2) = \sqrt{\pi}.$$

Select five out of following six problems.

1.(20pts.) A spaceship, moving away from the earth at a speed of $0.90c$, reports back by transmitting on a frequency (measured in the spaceship) of 100 MHz.

(a) Will the signal be red or blue shifted when it is received on the earth?

(b) What frequency the earth receiver must be tuned to for receiving this signal?

Solution: (a) The spaceship is receding from the earth. Therefore, the signal it emits will be red shifted when it is received on the earth.

$$(b) \quad \nu_o = \sqrt{\frac{1-\beta}{1+\beta}} \nu_e = \sqrt{\frac{1-0.9}{1+0.9}} 100 \text{ MHz} = 23 \text{ MHz}.$$

2. (20pts.) A particle moves on the xy -plane with a speed v at an angle θ with respect to the x -axis ($-\pi < \theta < \pi$, $\theta \neq 0$).

(a) At what speed v' will the particle be observed in the frame S' which is moving along the x -axis with the speed $u = v$.

(b) What is the corresponding angle θ' of the velocity of the particle in the frame S' ?

Solution: (a) The three components of the velocity of the particle in the S frame is

$$v_x = v \cos \theta, \quad v_y = v \sin \theta, \quad v_z = 0.$$

Then, by using the Lorentz transformation of velocity, we have

$$\begin{aligned} v'_x &= \frac{v_x - u}{1 - \beta v_x / c} = \frac{v \cos \theta - v}{1 - \beta v \cos \theta / c} = -\frac{1 - \cos \theta}{1 - \beta^2 \cos \theta} v, \\ v'_y &= \frac{v_y}{\gamma(1 - \beta v_x / c)} = \frac{v \sin \theta}{\gamma(1 - \beta v \cos \theta / c)} = \frac{\sin \theta}{\gamma(1 - \beta^2 \cos \theta)} v, \\ v'_z &= \frac{v_z}{\gamma(1 - \beta v_x / c)} = 0. \end{aligned}$$

Thus

$$\begin{aligned} v' &= \sqrt{v'^2_x + v'^2_y + v'^2_z} = \frac{\sqrt{1 - 2 \cos \theta + \cos^2 \theta + (1 - \beta^2) \sin^2 \theta}}{1 - \beta^2 \cos \theta} v \\ &= \frac{\sqrt{2 - 2 \cos \theta - \beta^2 \sin^2 \theta}}{1 - \beta^2 \cos \theta} v = 2 \frac{\sin(\theta/2) \sqrt{1 - \beta^2 \cos^2(\theta/2)}}{1 - \beta^2 \cos \theta} v. \end{aligned}$$

(b)

$$\begin{aligned} \text{tg}(\theta') &= \frac{v'_y}{v'_x} = -\frac{\sin \theta}{\gamma(1 - \cos \theta)} = -\sqrt{1 - \beta^2} \text{ctg}\left(\frac{\theta}{2}\right), \\ \theta' &= \begin{cases} \pi - \arctg\left(\sqrt{1 - \beta^2} \text{ctg}\left(\frac{\theta}{2}\right)\right), & 0 < \theta < \pi \\ -\pi - \arctg\left(\sqrt{1 - \beta^2} \text{ctg}\left(\frac{\theta}{2}\right)\right), & -\pi < \theta < 0. \end{cases} \end{aligned}$$

3. (20pts.) A gas of one mole satisfies the equation of state $p v = RT(1 + Bp)$ where p, v and T are the pressure, molar volume and temperature respectively, R the so-called gas constant, and B a function of T .

(a) Find the coefficient of thermal expansion $\alpha = 1/V (\partial V / \partial T)_p$ of the gas.

(b) Find the isothermal compressibility $\kappa = -1/V (\partial V / \partial p)_T$ of the gas.

(c) Show that both α and κ approach the values of the ideal gas as the pressure p tends to zero.

Solution: (a) Rewriting the equation of state as $v = RTp^{-1}(1 + Bp)$ and making logarithm of both sides of it, we have

$$\ln v = \ln T - \ln p + \ln(1 + Bp) + \ln R.$$

Thus

$$\alpha = \frac{1}{v} \left(\frac{\partial v}{\partial T} \right)_p = \left(\frac{\partial \ln v}{\partial T} \right)_p = \frac{1}{T} + \frac{p}{1 + Bp} \frac{dB}{dT}.$$

(b)

$$\kappa = -\frac{1}{v} \left(\frac{\partial v}{\partial p} \right)_T = -\left(\frac{\partial \ln v}{\partial p} \right)_T = \frac{1}{p} - \frac{B}{1 + Bp}.$$

(c) When $p \rightarrow 0$, obviously we have $\alpha \rightarrow \frac{1}{T}$ and $\kappa \rightarrow \frac{1}{p}$.

Notice that for the ideal gas, we have $v = \frac{RT}{p}$ or $\ln v = \ln T - \ln p + \ln R$. And

$$\alpha = \frac{1}{v} \left(\frac{\partial v}{\partial T} \right)_p = \frac{1}{T},$$

$$\kappa = -\frac{1}{v} \left(\frac{\partial v}{\partial p} \right)_T = \frac{1}{p}.$$

Therefore, both α and κ approach the values of the ideal gas as the pressure p tends to zero.

4. (20pts.) Consider an ideal gas. Suppose its total amount is one mole.

(a) Show that $ds = c_v dT/T + R dv/v$ where s, T and v are the molar entropy, temperature and molar volume of the ideal gas respectively, R the gas constant, c_v the molar heat capacity at constant volume.

(b) The ideal gas undergoes a (polytropic) process such that $p v^n = C$ where n and C are both constant ($n \geq 0, n \neq 1$). Find Δs , the change in molar entropy, when the system changes from the initial volume v_1 to the final volume v_2 in this process. Suppose c_v is a constant.

(c) Find the heat capacity c_n of the ideal gas for the process that is described in (b).

Solution: (a) For a mole of ideal gas, the fundamental equation of thermodynamics can be written as

$$du = Tds - pdv$$

where u is the molar internal energy and $du = \left(\frac{\partial u}{\partial T} \right)_v = c_v dT$. By using the equation of state $p v = RT$, we have

$$c_v dT = Tds - \frac{RT}{v} dv \quad \text{or} \quad ds = c_v \frac{dT}{T} + R \frac{dv}{v}. \quad (1)$$

(b) For the $p v^n = C$, we have

$$\ln p + n \ln v = \ln C.$$

Taking differentials of both sides, we have

$$\frac{dp}{p} + n \frac{dv}{v} = 0 \quad (2)$$

From the equation of state, we have $\ln p + \ln v = \ln T + \ln R$.

Taking differentials of both sides, we have

$$\frac{dp}{p} + \frac{dv}{v} = \frac{dT}{T} \quad (3)$$

From (2) and (3), we obtain

$$\frac{dT}{T} = (1-n) \frac{dv}{v} \quad (4)$$

Substituting (4) into (1), we have

$$ds = c_v \frac{dT}{T} + R \frac{dv}{v} = [(1-n)c_v + R] \frac{dv}{v}.$$

Thus

$$\Delta s = \int ds = \int_{v_1}^{v_2} [(1-n)c_v + R] \frac{dv}{v} = [(1-n)c_v + R] \ln \frac{v_2}{v_1}.$$

(c) Using (4), eq. (1) can be rewritten as

$$ds = c_v \frac{dT}{T} + \frac{R}{1-n} \frac{dT}{T} = \left(c_v + \frac{R}{1-n} \right) \frac{dT}{T}.$$

Thus

$$dq = Tds = \left(c_v + \frac{R}{1-n} \right) dT$$

and

$$c_n = c_v + \frac{R}{1-n}.$$

5. (20 pts.) Given a gas with the total number of particles fixed.

(a) Prove that $(\partial T / \partial p)_H = -1/C_p (\partial H / \partial p)_T$ where T , p and H are the temperature, pressure and enthalpy of the system respectively, and C_p the heat capacity at constant pressure (Hint: The partial derivative $(\partial H / \partial p)_T$ implies that we can take H as a function of T and p , i.e., $H = H(T, p)$).

(b) Show from $H = G + TS$ that $(\partial T / \partial p)_H = 1/C_p [T(\partial V / \partial T)_p - V]$ where G , S and V are the Gibbs free energy, entropy and volume of the system respectively.

(c) Find $(\partial T / \partial p)_H$ for the ideal gas.

Solution: (a) Since $H = H(T, p)$, then

$$dH = \left(\frac{\partial H}{\partial T} \right)_p dT + \left(\frac{\partial H}{\partial p} \right)_T dp = C_p dT + \left(\frac{\partial H}{\partial p} \right)_T dp.$$

The above equation can be rewrite as

$$dT = \frac{1}{C_p} dH - \frac{1}{C_p} \left(\frac{\partial H}{\partial p} \right)_T dp.$$

Therefore,

$$\left(\frac{\partial T}{\partial p} \right)_H = -\frac{1}{C_p} \left(\frac{\partial H}{\partial p} \right)_T.$$

(b) By using the Legend transformation $G = H - TS$, we have

$$\left(\frac{\partial T}{\partial p} \right)_H = -\frac{1}{C_p} \left(\frac{\partial H}{\partial p} \right)_T = -\frac{1}{C_p} \left[\left(\frac{\partial G}{\partial p} \right)_T + T \left(\frac{\partial S}{\partial p} \right)_T \right].$$

From $dG = d(U - TS + pV) = -SdT + Vdp$, we have

$$V = \left(\frac{\partial G}{\partial p} \right)_T, \quad S = -\left(\frac{\partial G}{\partial T} \right)_p.$$

Therefore

$$\begin{aligned}\left(\frac{\partial T}{\partial p}\right)_H &= -\frac{1}{C_p} \left\{ V - T \left[\frac{\partial}{\partial p} \left(\frac{\partial G}{\partial T} \right)_p \right]_T \right\} = \frac{1}{C_p} \left\{ T \left[\frac{\partial}{\partial T} \left(\frac{\partial G}{\partial p} \right)_T \right]_p - V \right\} \\ &= \frac{1}{C_p} \left[T \left(\frac{\partial V}{\partial T} \right)_p - V \right].\end{aligned}$$

(c) For an ideal gas, the equation of state gives $V = \frac{Nk_B T}{p}$, where N is the number of particles. Thus

$$\left(\frac{\partial V}{\partial T}\right)_p = \frac{Nk_B}{p} = \frac{V}{T}.$$

Therefore

$$\left(\frac{\partial T}{\partial p}\right)_H = \frac{1}{C_p} \left[T \left(\frac{\partial V}{\partial T} \right)_p - V \right] = \frac{1}{C_p} \left[T \frac{V}{T} - V \right] = 0.$$

6. (20pts.) The Maxwell speed distribution has the form $dn = C \exp(-mv^2/2k_B T) v^2 dv$ where m denotes the mass of the molecule of the gas, T the temperature, and k_B the Boltzmann constant. The coefficient C is a constant. We denote $\varepsilon = (1/2)mv^2$ as the energy of the molecule of the gas.

(a) Calculate the statistical average of the speed v , i.e., $\langle v \rangle$.

(b) Calculate $\langle v^2 \rangle$ and $\langle \varepsilon \rangle$.

(c) Calculate $\langle \varepsilon^2 \rangle$.

(d) The uncertainty (or fluctuation) ΔO of a quantity O is defined as $\Delta O = \sqrt{\langle O^2 \rangle - \langle O \rangle^2}$. Find the uncertainties Δv and $\Delta \varepsilon$.

Solution: (a)

$$\begin{aligned}\langle v \rangle &= \frac{\int v dn}{\int dn} = \frac{\int_0^{+\infty} \exp\left(-\frac{mv^2}{2k_B T}\right) v^3 dv}{\int_0^{+\infty} \exp\left(-\frac{mv^2}{2k_B T}\right) v^2 dv} = \sqrt{\frac{2k_B T}{m}} \frac{\int_0^{+\infty} \exp(-x^2) x^3 dx}{\int_0^{+\infty} \exp(-x^2) x^2 dx} \\ &= \sqrt{\frac{2k_B T}{m}} \frac{\int_0^{+\infty} \exp(-y) y dy}{\int_0^{+\infty} \exp(-y) y^{\frac{1}{2}} dy} = \sqrt{\frac{2k_B T}{m}} \frac{\Gamma(2)}{\Gamma(3/2)} = \sqrt{\frac{2k_B T}{m}} \frac{1}{\frac{1}{2}\sqrt{\pi}} = \sqrt{\frac{8k_B T}{\pi m}}.\end{aligned}$$

(b)

$$\begin{aligned}\langle v^2 \rangle &= \frac{\int v^2 dn}{\int dn} = \frac{\int_0^{+\infty} \exp\left(-\frac{mv^2}{2k_B T}\right) v^4 dv}{\int_0^{+\infty} \exp\left(-\frac{mv^2}{2k_B T}\right) v^2 dv} = \frac{2k_B T}{m} \frac{\int_0^{+\infty} \exp(-x^2) x^4 dx}{\int_0^{+\infty} \exp(-x^2) x^2 dx} \\ &= \frac{2k_B T}{m} \frac{\int_0^{+\infty} \exp(-y) y^{\frac{3}{2}} dy}{\int_0^{+\infty} \exp(-y) y^{\frac{1}{2}} dy} = \frac{2k_B T}{m} \frac{\Gamma(5/2)}{\Gamma(3/2)} = \frac{3k_B T}{m}.\end{aligned}$$

$$\langle \varepsilon \rangle = \frac{\int \frac{1}{2}mv^2 dn}{\int dn} = \frac{1}{2}m \frac{\int v^2 dn}{\int dn} = \frac{1}{2}m \langle v^2 \rangle = \frac{3}{2}k_B T.$$

(c)

$$\begin{aligned}
\langle \varepsilon^2 \rangle &= \frac{\int \frac{1}{4} m^2 v^4 dn}{\int dn} = \frac{1}{4} m^2 \frac{\int_0^{+\infty} \exp\left(-\frac{mv^2}{2k_B T}\right) v^6 dv}{\int_0^{+\infty} \exp\left(-\frac{mv^2}{2k_B T}\right) v^2 dv} = \frac{1}{4} m^2 \left(\frac{2k_B T}{m}\right)^2 \frac{\int_0^{+\infty} \exp(-x^2) x^6 dx}{\int_0^{+\infty} \exp(-x^2) x^2 dx} \\
&= (k_B T)^2 \frac{\int_0^{+\infty} \exp(-y) y^{\frac{5}{2}} dy}{\int_0^{+\infty} \exp(-y) y^{\frac{1}{2}} dy} = (k_B T)^2 \frac{\Gamma(7/2)}{\Gamma(3/2)} = \frac{15}{4} (k_B T)^2.
\end{aligned}$$

(d)

$$\begin{aligned}
\Delta v &= \sqrt{\langle v^2 \rangle - \langle v \rangle^2} = \sqrt{\frac{3k_B T}{m} - \frac{8k_B T}{\pi m}} = \sqrt{\frac{(3\pi - 8)k_B T}{\pi m}}. \\
\Delta \varepsilon &= \sqrt{\langle \varepsilon^2 \rangle - \langle \varepsilon \rangle^2} = \sqrt{\frac{15}{4} (k_B T)^2 - \frac{9}{4} (k_B T)^2} = \sqrt{\frac{3}{2}} k_B T.
\end{aligned}$$